FORGING IMAGE WATERMARKS BY SCHEME IDENTIFICATION AND NATIVE RE-ENCODING

Muhammad Saad Haroon
Matriculation No. 7088620
Department of Computer Science
Universität des Saarlandes
muha00003@stud.uni-saarland.de

Jayasankar Kumar Santhirani
Matriculation No. 7089851
Department of Computer Science
Universität des Saarlandes
jaku00013@stud.uni-saarland.de

Team V • Code: https://github.com/SaadH-077/TML-Watermark-Forging

1 INTRODUCTION

We are given 8 unknown image watermarking schemes $\mathrm{WM}_1, \ldots, \mathrm{WM}_8$. For each scheme we receive 25 watermarked "carrier" images that all carry *one* shared hidden message, and we must transplant that watermark onto 25 assigned clean target images so that a hidden detector recovers the message while each forged image stays perceptually close to its target. The carriers and the targets are different images, so this is true transplantation rather than re-stamping the same picture, and the resolution is fixed per scheme. A submission is a flat archive of 200 PNG files, one per target.

The score is $S_{\text{final}} = S_{\text{det}} \cdot S_{\text{qlt}}$, with $S_{\text{det}} = \max\big(0,\, 2(\text{bitacc} - 0.5)\big)$ and $S_{\text{qlt}} = \exp(-8\,\text{LPIPS})$. Two properties of this score decide the whole design. First, S_{det} is a hard gate: a bit-accuracy of 0.5 sets $S_{\text{det}} = 0$ and zeroes the image no matter how good it looks. Second, LPIPS is measurable locally, but bit-accuracy is not, because we hold no detector. The only detection signal is the leaderboard, which allows one submission per hour, keeps the best result, and scores on a 30% public subset. The practical consequence is that detection strength, not visual quality, is the binding constraint, so our rule was to land the bits first and recover LPIPS second. Our final submission reaches a public score of **0.8364**.

2 APPROACH

Identification beats blind transplant. The averaging attack from the watermarking literature estimates the watermark as the mean carrier residual and adds it to the target [1]. This fails on frequency-domain and content-adaptive marks. On the first two schemes where we could also decode, the difference was clear: a native re-encode reached $S_{\text{final}} = 0.89$ (WM_1) and 0.86 (WM_2), while an additive transplant scored close to 0. The better route is to *identify* the scheme, decode the shared message from its carriers, and re-embed that message into the target with the scheme's own genuine encoder. The forged image is then a real watermark of that scheme, so it decodes at bit-accuracy 1.0 and generalizes across the public and private split. Identification became our main strategy.

The oracle test. Because the 25 carriers share one message, we can identify a scheme without any detector. For a candidate public decoder we decode all 25 carriers: if they agree on a single message (carrier-agreement near 1.0) while clean images decode randomly (clean-agreement near 0.5), the scheme is identified. We accepted a match only with a *validated positive control*, a genuine encode then decode round trip proving the decoder reads real marks of that scheme, and a known-negative control from an already identified scheme. This guarding mattered, because several apparent matches were artifacts. A common failure was a decoder that returns a constant output regardless of its input; another was content correlation that also lifts clean images. As one example, running classical decoders on WM_3's chroma channels reports carrier-agreement 1.000, but the same decoder returns the identical constant bits on pure random noise, which is a degenerate reading and not a recovered message.

What we did for each watermark. We identified seven of the eight schemes and re-encoded each one natively at round-trip bit-accuracy 1.0 (Table 1). WM_1 is `dwtDct` from the `invisible-watermark` library, a 30-bit DWT-DCT mark whose message decodes as all ones; we re-embed with the genuine encoder and then amplify the residual at strength $s = 1.5$ to widen the quantization margin, which the robustness audit below motivates. WM_2 is RivaGAN [2], a 32-bit attention-based mark whose encoder sets the quality floor among the natives at LPIPS 0.016. WM_4

is VINE-R [7], re-encoded with the released generative-prior encoder at its native 256^2. WM_5 is CIN [4] and WM_6 is MBRS [3], two learned encoder-decoder marks that we re-embed at their native 128^2 and 256^2. WM_7 and WM_8 are two TrustMark variants [6] at 512^2; WM_8 matched only after we tried variant P with error correction disabled, which is why an earlier sweep of variant Q alone missed it. WM_3 is the one scheme without a public decoder and is handled separately below.

Table 1: Per-scheme forge. The seven identified schemes are genuine native re-encodes (bit-accuracy 1.0); WM_3 is the only transplant. LPIPS and S_{final} are per-scheme local estimates.

WM	Scheme (message)	Forge	LPIPS / S_{final}
WM_1	dwtDct, 30-bit	native re-encode, residual amplified s=1.5	0.006 / 0.89
WM_2	RivaGAN, 32-bit	native re-encode	0.016 / 0.86
WM_3	custom chroma (unidentified)	color-axis $R{+}B{-}2G$ transplant	0.023 / 0.43
WM_4	VINE-R, 100-bit	native re-encode (256^2)	0.003 / 0.94
WM_5	CIN, 30-bit	native re-encode (128^2)	0.001 / 0.99
WM_6	MBRS, 256-bit	native re-encode (256^2)	0.002 / 0.96
WM_7	TrustMark-Q, 61-bit	native re-encode (512^2)	0.001 / 0.99
WM_8	TrustMark-P (no ECC), 100-bit	native re-encode (512^2)	0.001 / 0.99

WM_3: a custom scheme with no public decoder. For WM_3 we ruled out about 27 public decoders under the positive-control bar, including all VINE W-Bench baselines, RoSteALS, InvisMark, CR-Mark, Stable Signature [5], LaWa, SepMark, PIMoG, EditGuard, and classical and spread-spectrum families. From-scratch forensics place WM_3 in a low-frequency, green-magenta chroma band that is content-adaptive and content-keyed. A Craver-style diagnostic [1] makes the last point concrete: a norm-matched transplant of the averaged template reaches bit-accuracy 0.88 on the analogous VINE mark, near its optimum, but WM_3's detector accepts the same-quality template at only about 0.72. So WM_3 has no public decoder to re-encode with, and its per-image key limits any fixed-template transplant. Our best forge is a color-axis ($R{+}B{-}2G$) chroma transplant with a bilateral-denoised template, capped near $S_{\text{final}} \approx 0.43$. We confirmed the ceiling three ways: a strength sweep plateaus, a robust estimator (a 20% trimmed mean) that helps the decodable analogs does not transfer to WM_3, and a learned diffusion forger stays below transplant parity, held back by the 25-carrier data limit.

Native robustness locates the residual gap. The leaderboard grades bit-accuracy after an unknown pre-detection transform, so clean-PNG bit-accuracy overestimates the server. We built a local robustness oracle that decodes a forge, applies JPEG, resizing, noise, and blur, then decodes again and measures the surviving bits. Six of the seven natives are already robust and near optimal. Only WM_1, the classical dwtDct mark, is fragile, with surviving bit-accuracy 0.60 that collapses under JPEG. A leaderboard probe that strengthened WM_1's quantization margin was a wash of $+0.0002$: the quality cost cancelled the detection gain, and dwtDct is JPEG-fragile at every strength. The natives therefore sit near their achievable ceiling under these public encoders, which is an informative negative result rather than a missed lever.

3 Results

Table 2 shows the public leaderboard progression. Every gain came from a per-scheme method that generalizes across the public and private split, not from tuning to the public sample. Identifying CIN and MBRS added 0.057; identifying VINE-R and TrustMark-P added another 0.10; the two WM_3 steps and the robustness work added the rest. Decomposing the total, the seven native schemes average about $0.97 \cdot S_{\text{qlt}}$ each while WM_3 contributes about 0.43, so that single scheme is almost the entire remaining gap to the strongest teams near 0.90. The best submission rebuilds from the repository with one command (`build_best.py`), and every reported number has a script behind it.

Table 2: Public leaderboard score for each submitted configuration. Schemes are added and refined one change at a time so each step is attributable.

Run	Configuration	Score
1	3 native (WM_1, WM_2, WM_7) + transplant elsewhere	0.6613
2	+ CIN, MBRS native (5 native)	0.7188
3	+ VINE-R, TrustMark-P native (7 native) + WM_3 residual transplant	0.8211
4	WM_3 in-band chroma transplant	0.8268
5	+ perceptually-masked natives	0.8274
6	WM_3 color-axis ($R+B-2G$) transplant	0.8362
7	**+ WM_1 residual amplification** $s=1.5$	**0.8364**

4 Conclusion

Watermark forgery under near black-box conditions is practical and cheap. For any deployed public scheme, an attacker who collects a handful of same-message watermarked images can identify the scheme and re-encode with its own public encoder, producing a genuine and generalizing forgery that stamps arbitrary content as authentically watermarked and defeats provenance claims. Even unknown learned marks are partially forgeable by averaging. The one scheme we could not forge to native quality, WM_3, is exactly the one that is content-keyed with a non-public detector. That is the defensive lesson: reliable provenance needs secret, content-bound, keyed watermarks whose detectors reject transplanted marks, not merely imperceptible marks built on public or guessable encoders.

References

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